

QoE Characterization for Video-on-Demand Services in 4G WiMAX Networks

Amitabha Ghosh (USC), Malolan Chetlur (IRL), Partha Dutta (IRL), Umamaheshwari Devi (IRL), Shivkumar Kalyanaraman (IRL)
Next Generation Telecom Research, IBM India Research Laboratory, Bangalore

Motivation

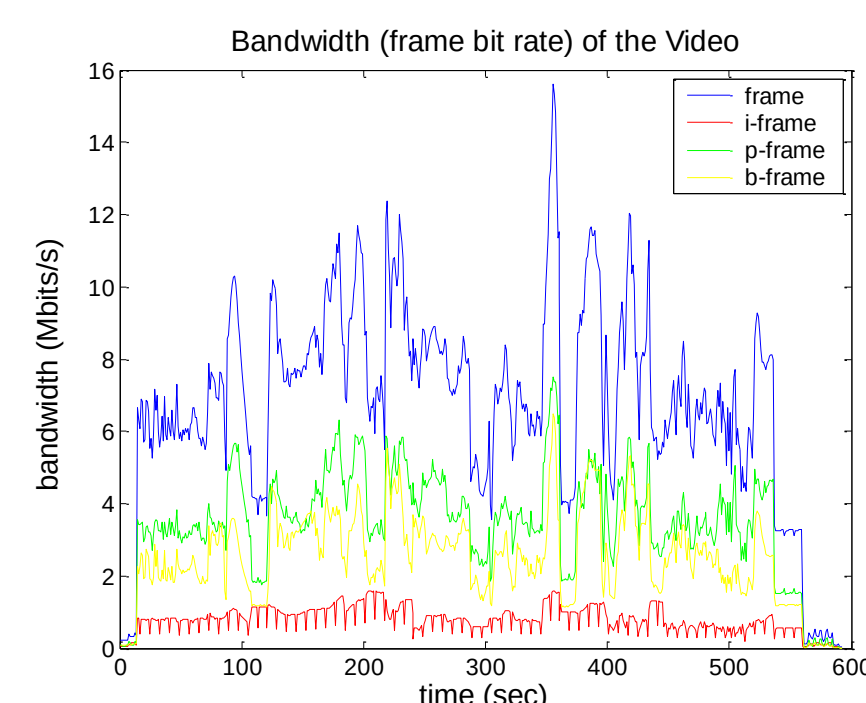
- ❑ Active deployment of *triple* and *quadruple services* by network operators delivering voice, real-time high-quality video, and value-added data services over converged networks (e.g., 4G).
- ❑ Need to guarantee subscribers' *Quality-of-Experience (QoE)* and provide differentiated services in the face of heterogeneous end devices, varying wireless channels, resource constraints, etc.
- ❑ Lack of *QoE orchestration* models that map network events (e.g., variation in bandwidth, delay, error rates, jitter) to subscribers' QoE.

State-of-the-Art in QoE

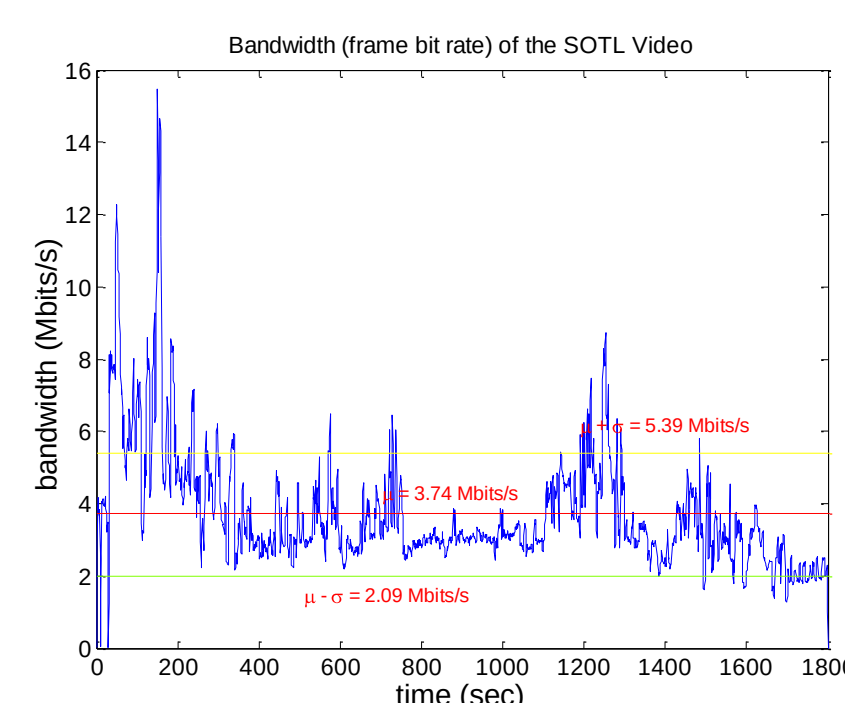
- ❑ **Video Quality Attributes:** Session quality (initial buffering, re-buffering, audio video synchronization), video quality (frame quality, frame-rate, stalling), and nature of the content contribute to user' QoE.
- ❑ Several unreliable objective (e.g., *PSNR*, *SSIM*) and subjective metrics (e.g., *MOS*, *VQM*) to measure QoE
- ❑ **Current Solutions:** Bit-rate switching, feedback-based encoding, etc.
- ❑ Very little understanding of the major causes of user perceived degradation.

Evaluation Video Characteristics

- ❑ Sony Video
 - ❖ MPEG-4, VBR, frame size CIF 352 x 288
 - ❖ Video duration: 589 sec, 7681 frames, GoP size: 16
 - ❖ Average BW: 6.727 Mbits/sec, SD BW: 2.8 Mbits/sec
- ❑ Silence of the Lambs
 - ❖ MPEG-4 VBR, frame size CIF 352 x 288
 - ❖ Video duration: 1800 sec, 53953 frames, GoP size: 16
 - ❖ Average BW: 3.74 Mbits/sec, SD BW: 1.75 Mbits/sec



Sony Video: variation of bit rate with time of different frames

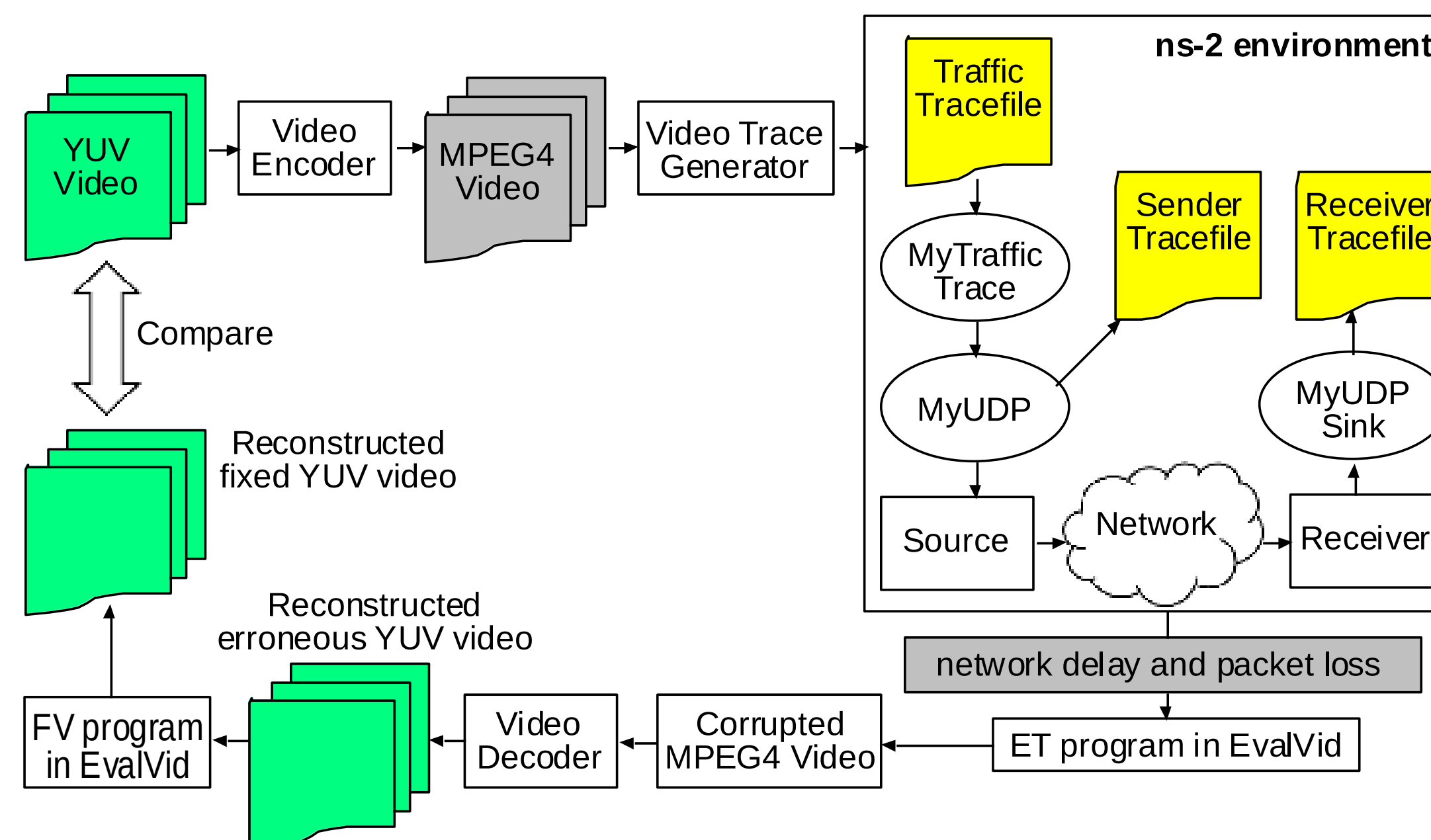


Silence of the Lambs: variation of bit rate with time

Problem Statement

- ❑ To build robust QoE models by measurement and detection of streaming **Video-on-Demand (VoD)** characteristics and network parameters over 4G WiMAX networks.
- ❑ To propose orchestration mechanisms using the above QoE model

Simulation Framework & Methodology



Business Context

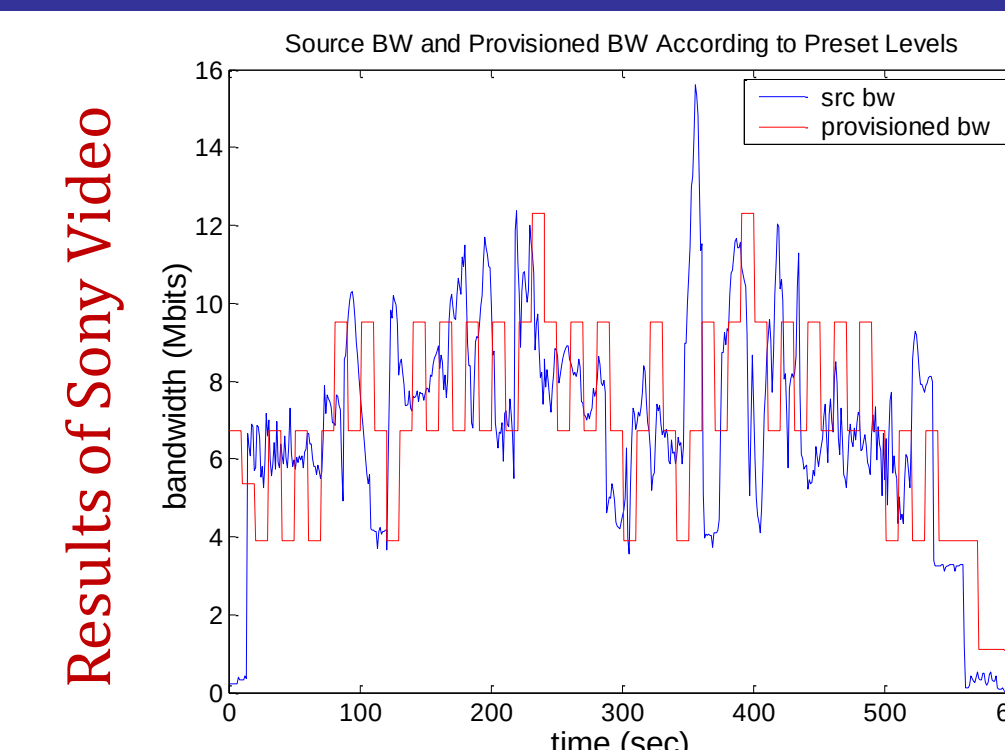
- ❑ Extending IBM's business model by using new edge-based network appliances (e.g., **Services Substrate Appliance - SSA**) into key infrastructure elements at various entry points (e.g., gateways, base stations)

- ❑ ns-2 combined with EvalVid
- ❑ Three new ns-2 agents
 - ❖ MyTrafficTrace
 - ❖ MyUDP
 - ❖ MyUDPSink
- ❑ Stream encoded VBR video traffic over a simple WiMAX network, and measure:
 - ❖ number of stalls
 - ❖ duration and distribution of stalls
 - ❖ receiver buffer content
- for different network parameters (e.g., BW)
- ❑ Generate received video and compare with source video

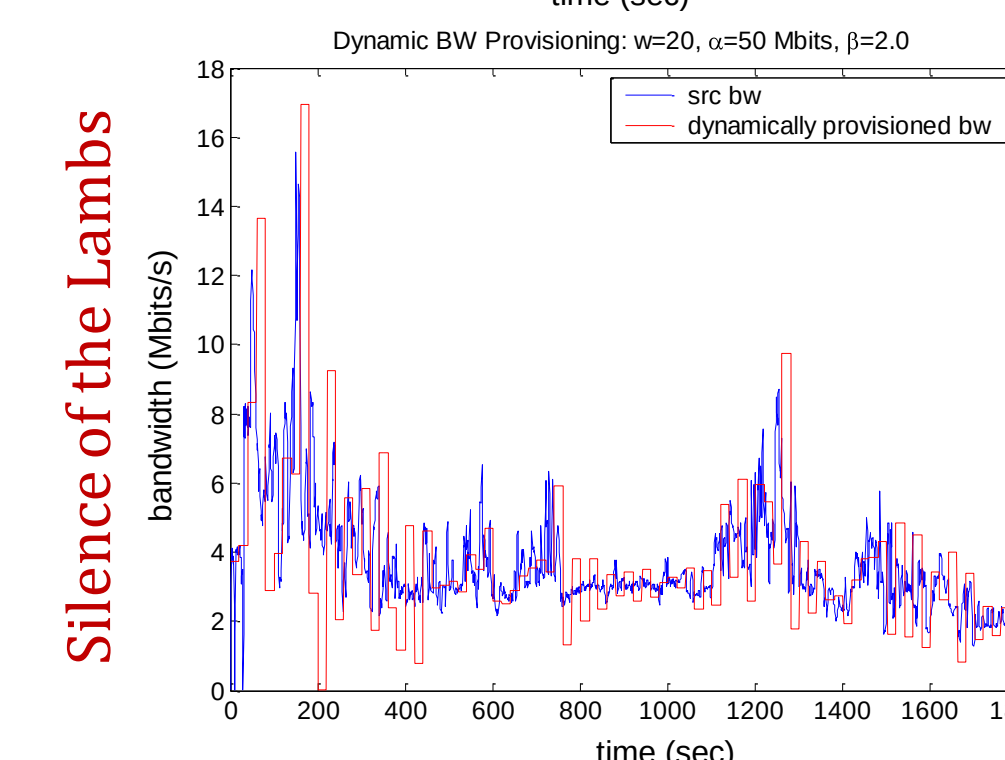
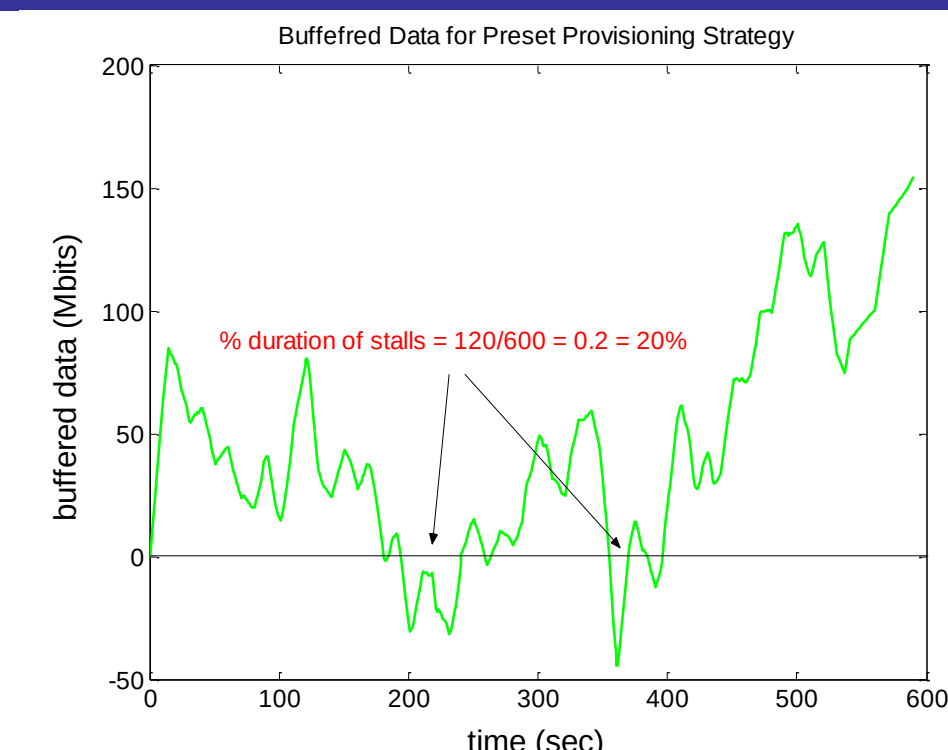
Bandwidth Provisioning Strategy & Evaluation Results

- ❑ **Fixed BW Provisioning**
 - ❖ Preset BW levels based on the source bit rate
 - ❖ Provision these bandwidths at fixed intervals depending on whether the current bit rate falls below or rises above the provisioned BW
- ❑ **Dynamic BW Provisioning**
 - ❖ Track cumulative moving average $\mu(i)$ and cumulative moving standard deviation $\sigma(i)$ of the source bit rate
 - ❖ Provision BW at fixed intervals such that
 - Provisioned BW for the next interval can withstand a variation of k times the SD of bit rate of the current interval
 - Estimated receiver buffer does not fall below a certain threshold α

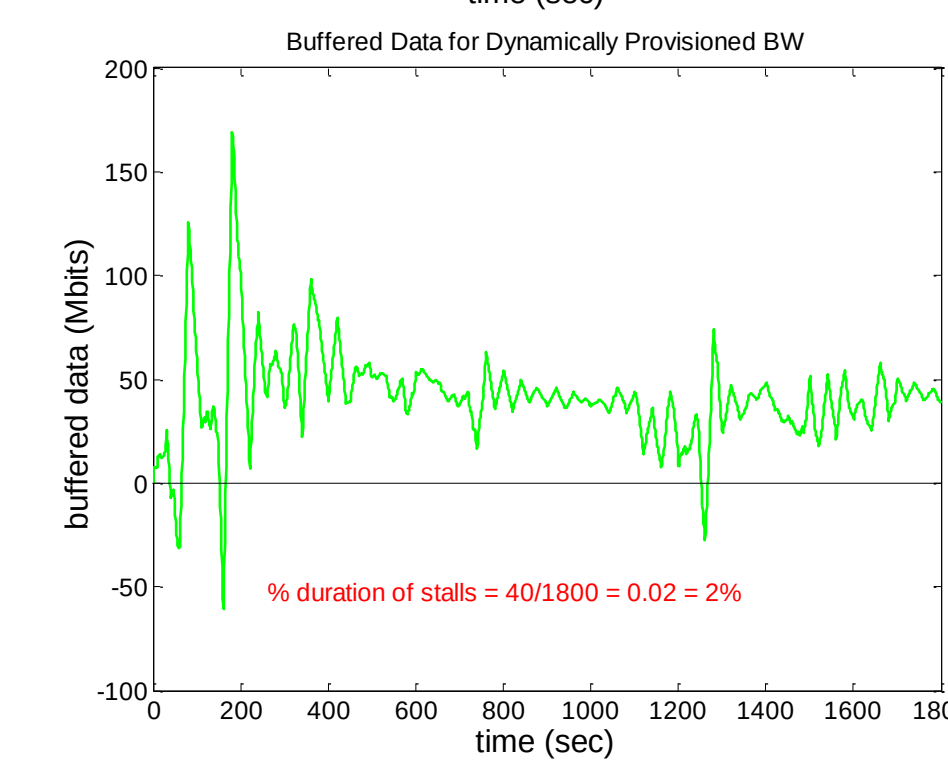
$$BW(i+1) = \mu(i) + k.\sigma(i) + \alpha/w - \beta.[\sum_j w.BW(j) - w.i.\mu(i)]/w$$



Results of Sony Video



Silence of the Lambs



References

- [1] Klaue, B. Rathke, and A. Wolisz, "EvalVid - A Framework for Video Transmission and Quality Evaluation", *Conference on Modeling Techniques and Tools for Computer Performance Evaluation*, September 2003.